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LLM Evaluation

Introduction





Why Evaluate LLMs?

LLM Evaluation



Why do we need Evaluation

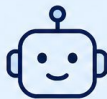
LLM Evaluation

Imagine asking two LLMs the same question:



Question: "What is the capital of France?"

LLM A



- ✓ Answers confidently.
- ✓ Sounds fluent.

LLM B



- ✓ Answers accurately.
- ✓ Explains its reasoning.



Sound
confident

Actually
correct

Which one is better?

Without evaluation, it's impossible to know



What is LLM Evaluation?

LLM Evaluation

LLM Evaluation is the process of measuring **how well a language model performs** on different tasks.

Why?

Evaluation helps us answer questions like:

- Is the answer correct?
- Is it relevant?
- Is it safe?
- Is it consistent?
- Is it useful?

When?

Evaluate an LLM:

- Before deployment
- After changing prompts
- After fine-tuning
- When comparing multiple models
- Continuously in production

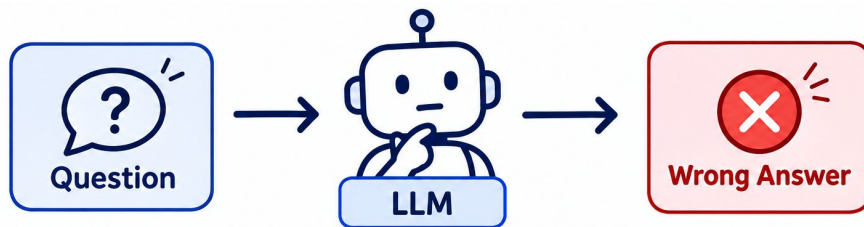


What can go wrong?

LLM Evaluation

Even powerful LLMs can:

- Hallucinate facts
- Misunderstand questions
- Give irrelevant answers
- Produce unsafe content
- Be inconsistent





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What should we Evaluate?

LLM Evaluation



Four Dimensions of LLM Evaluation

LLM Evaluation

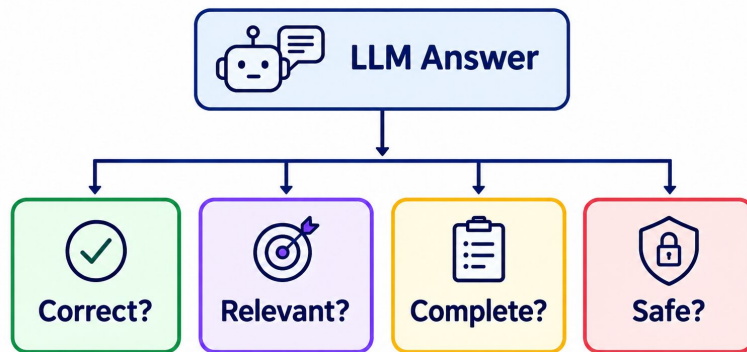
Every answer can be judged on four questions:

Is it Correct?

Is it Relevant?

Is it Complete?

Is it Safe?

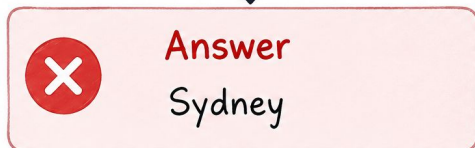




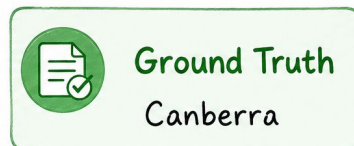
Correctness

LLM Evaluation

Does the answer match the expected or factual answer?



Incorrect



Correct

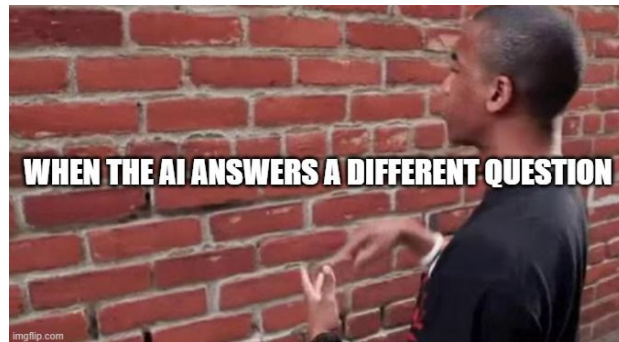
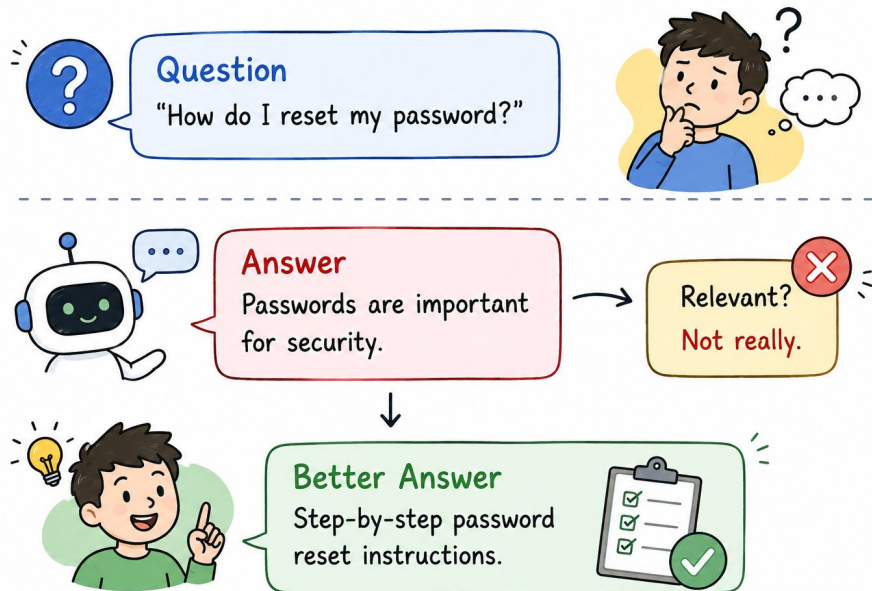




Relevance

LLM Evaluation

Does the answer actually answer the user's question?

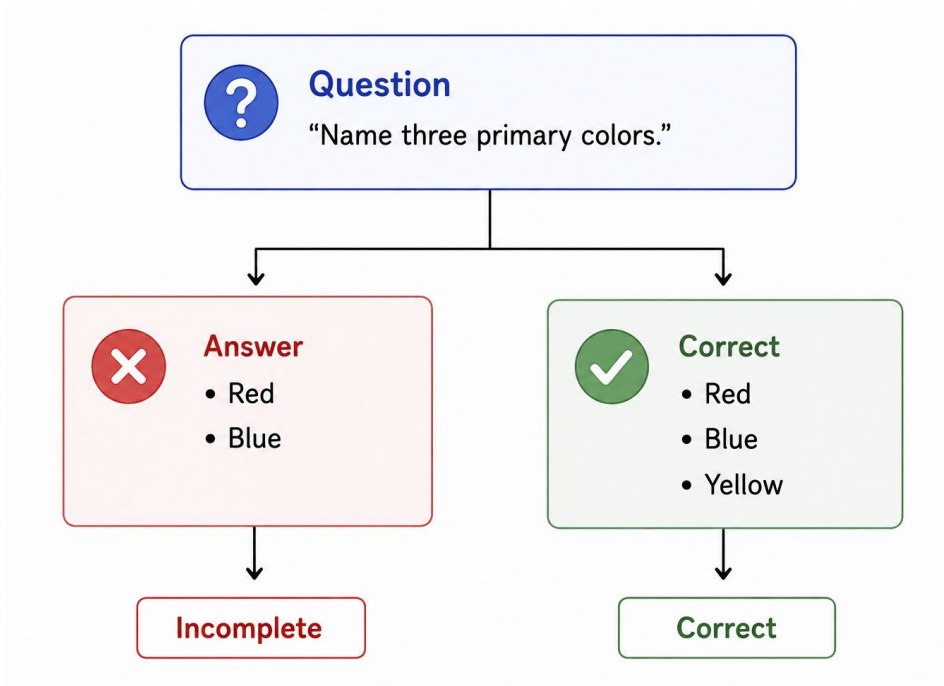




Completeness

LLM Evaluation

Did the model include all important information?





Safety

LLM Evaluation

Does the response avoid harmful or inappropriate content?

Examples

- Hate speech
- Dangerous advice
- Personal information leakage

Responsible AI must be both useful **and** safe.





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Human Vs Automatic Evaluation

LLM Evaluation



Human Evaluation

LLM Evaluation

Humans judge responses based on:

- Correctness
- Helpfulness
- Fluency
- Safety



Advantages

High Quality

Disadvantages

- Slow
- Expensive



Automatic Evaluation

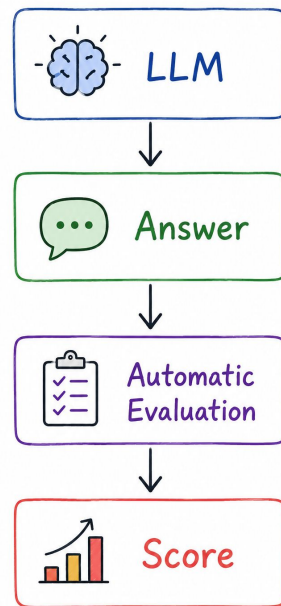
LLM Evaluation

Instead of humans, we can use:

- Benchmarks
- LLM-as-a-Judge
- Automated metrics

Advantages

- Fast
- Scalable



Disadvantages

May miss nuances



Human Vs Automatic

LLM Evaluation

Human	Automatic
Accurate	Fast
Expensive	Cheap
Slow	Scalable
Subjective	Consistent



When should we use Each?

LLM Evaluation

HUMAN

Best for:

- Final testing
- UX
- Safety review

AUTOMATIC

Best for:

- Development
- Regression testing
- Large datasets



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Common Evaluation Metrics

LLM Evaluation



Common Metrics Overview

LLM Evaluation

A good response should generally be:

- **Correct:** Contains accurate information.
- **Relevant:** Directly addresses the user's question.
- **Complete:** Includes the important information needed to answer.
- **Coherent:** Ideas are logically connected and easy to follow.
- **Fluent:** Uses natural, understandable language.
- **Safe:** Does not produce harmful or inappropriate content.



Sounds
Impressive

Is
actually
correct



Evaluation Criteria

LLM Evaluation

We can turn these qualities into questions that we can actually evaluate.

Quality	Evaluation Question
Correctness	Is the information factually correct?
Relevance	Does it answer the user's question?
Completeness	Did it include the important information?
Coherence	Do the ideas flow logically?
Fluency	Is the language natural and understandable?
Safety	Does the response avoid harmful content?



Evaluation Criteria

LLM Evaluation

Imagine an answer that

- ✓ Uses perfect grammar
- ✓ Is easy to read
- ✓ And confidently explains something that is **completely false**.



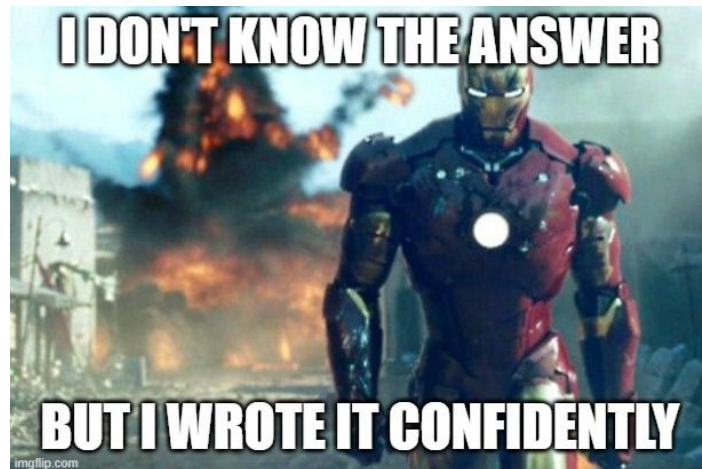
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| Which criterion failed?



A

| Answer: Correctness'





A Major LLM Problem: Hallucination

LLM Evaluation

A hallucination occurs when an LLM generates information that is **false**, **unsupported**, or **made up**, while presenting it as if it were true.

 User:	Who wrote the book <i>The Great Gatsby</i> ?
 LLM:	Ernest Hemingway wrote <i>The Great Gatsby</i> .

Evaluation

- Fluent? 
- Relevant? 
- Confident? 
- Correct? 

The answer sound reasonable but is wrong



How to detect Hallucinations

LLM Evaluation

Once hallucination becomes an evaluation concern, we need to ask:

How can we determine whether an answer is actually supported?

Reference

Canberra is the capital of Australia.

Generated Answer

Sydney is the capital of Australia.

Comparison



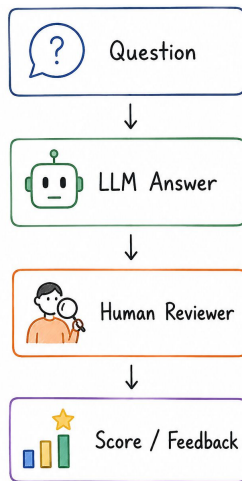
Incorrect



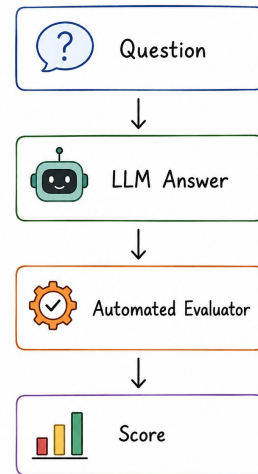
Who can Evaluate an LLM?

LLM Evaluation

Human Evaluation



LLM Evaluation





Why use both?

LLM Evaluation

Human Evaluation

- Better at understanding nuance
- Can judge usefulness and quality
- Takes more time and effort

LLM Evaluation

- Faster
- Can evaluate many responses
- Useful for repeated testing
- May miss subtle issues

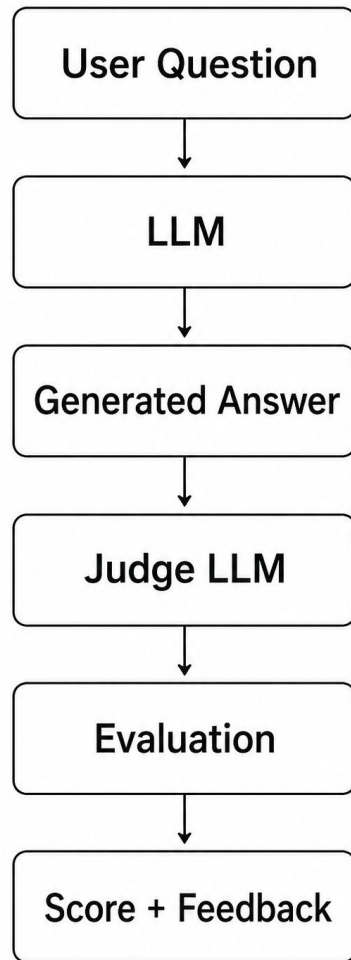


LLM-as-a-Judge

LLM Evaluation

Instead of asking a human to inspect every response, we give the response to an evaluation model and ask it to assess specific criteria.

LLM-as-a-Judge uses an LLM to evaluate another LLM's response.





LLM-as-a-Judge

LLM Evaluation

Example



Question:

Explain photosynthesis in simple terms.



Generated Answer:

Plants use sunlight to make food from water and carbon dioxide.



Judge LLM:

- | | |
|----------------|-----|
| • Correctness: | 5/5 |
| • Relevance: | 5/5 |
| • Clarity: | 4/5 |



When is it Useful?

LLM as a Judge

LLM-as-a-Judge is useful when:

- There are many responses to evaluate.
- The task is difficult to evaluate with simple exact-match metrics.
- We want scalable evaluation during development.



<https://towardsdatascience.com/llm-as-a-judge-a-practical-guide/>



Limitations

LLM as a Judge

A judge LLM is **not automatically perfect**.

- It can also make mistakes or show biases.

So automated evaluation should be validated against **human judgment**, especially for important applications.





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Benchmarking and Continuous Evaluation

LLM Evaluation



Benchmarks

LLM Evaluation

- MMLU
- GSM8K
- HumanEval
- TruthfulQA

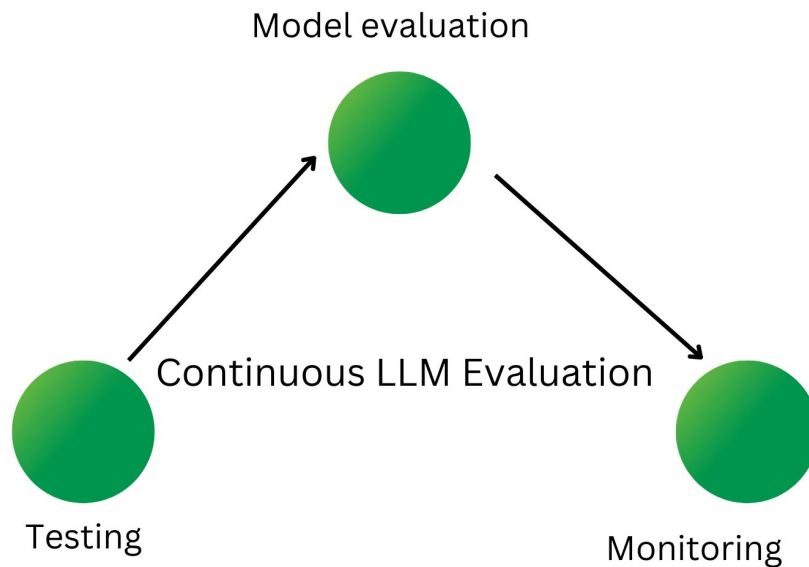
Different benchmarks measure different abilities.



Continuous

LLM Evaluation

Evaluation isn't done once.





Improving LLM Performance

LLM Evaluation

If scores are low, consider improving:

- Prompt
- Retrieval (RAG)
- Fine-tuning
- Training data
- Safety guardrails



Summary

- Why LLMs must be evaluated
- Human vs automatic evaluation
- Common quality metrics
- Hallucinations
- LLM-as-a-Judge
- Continuous evaluation



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Thank You

We'd love to hear from you. If you have any questions,
feel free to reach out to our team anytime!